sitemix: An R Package for Site- and Group-Level Proportions, Rates, and Sampling Uncertainty from Individual and Aggregate Administrative Data

JoonHo Lee and Alison Hooper

Abstract. Proportions and rates computed from administrative records carry much of the routine comparison of schools, clinics, and other program sites: they enter funding formulas, accountability reports, and public rankings. Survey estimates travel with margins of error, and an official handbook tells data users how to read them; administrative rates almost never do. One reason, we believe, is that the computation behind a defensible margin of error is less straightforward than it appears. Site-level panels mix overlapping indicators, whose joint sampling covariance any combined analysis requires, with boundary cells, publisher suppression, and, in many agencies, aggregate tables in place of individual records. The required closed-form results exist but are scattered across a century of statistical literature. We describe sitemix, an R package that assembles these results behind a single interface. Individual records, sufficient counts, and published tables all yield the same schema-stable table of estimates, standard errors, and, where identified, joint covariance; where the data cannot identify a quantity, the package reports sharp Fréchet bounds and labeled stress scenarios in place of a silent assumption. Diagnostics and suppression audits keep long-maintained reporting pipelines auditable, and the output feeds empirical Bayes, meta-analytic, and small-area methods without further preparation. In a worked example on a simulated pre-kindergarten panel, attaching uncertainty reorders a site ranking, and two of the ten highest-ranked sites leave the top ten. The package makes margins of error as available for administrative rates as they already are for surveys.

Keywords: sampling uncertainty; administrative data; proportions and rates; empirical Bayes; multisite programs; R

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Lee: Department of Educational Studies in Psychology, Research Methodology, and Counseling, The University of Alabama, Tuscaloosa, AL, USA. jlee296@ua.edu. Hooper: Department of Curriculum and Instruction, The University of Alabama, Tuscaloosa, AL, USA. alhooper2@ua.edu.

1. **Introduction**

Administrative records are increasingly used to monitor multisite programs at the level of the individual site. School systems report, for each campus, the proportion of students who are chronically absent or eligible for free or reduced-price meals (FRPM); pre-kindergarten programs publish, for each site, the proportion of children whose families participate in SNAP, WIC, or TANF; and health agencies release clinic-level counts of service uptake. These proportions and rates enter funding formulas, accountability reports, and program targeting. In routine practice they are computed as simple ratios of counts and are reported without any accompanying measure of sampling uncertainty, although the number of children per site is often small and varies widely between sites. In state pre-kindergarten programs this variation follows the number of funded classrooms at a site: a small rural school with one classroom may serve nine children, while a larger school with six classrooms may serve more than a hundred. An estimate based on nine children then enters rankings and policy discussions on the same basis as an estimate based on a hundred.

For survey estimates the analogous problem was solved long ago by convention. Estimates from the American Community Survey are published together with their margins of error, and an official handbook instructs data users in their interpretation (U.S. Census Bureau, 2020). No comparable convention exists for statistics computed from administrative counts. One reason, we believe, is that the required computation is less straightforward than it first appears. Site-level panels commonly involve

(a) overlapping binary indicators whose joint sampling covariance is needed for any combined analysis, since a child may receive both FRPM and SNAP;
(b) mutually exclusive categories on a simplex;
(c) boundary cells in which every child, or none, has the attribute, together with publisher suppression of small cells; and
(d) frequently, no individual-level data at all, since many agencies release only aggregate tables of counts.

Closed-form uncertainty results are available for each of these situations, but they are scattered across a century of statistical literature (e.g., Agresti, 2013; Anscombe, 1948; Cochran, 1977; Fréchet, 1951; Wilson, 1927; Zellner, 1962), and we are not aware of software that assembles them into a single implementation for the site $\times$ year $\times$ indicator panels that analysts actually hold.

The consequences extend beyond reporting, because site-level summaries are increasingly consumed by methods that treat the first-stage standard errors as known. Empirical Bayes (EB) shrinkage requires a standard error for each site in order to separate signal from noise (Efron, 2016; Fay & Herriot, 1979; Morris, 1983), and Walters (2024) presents it as a standard tool for institutional comparisons in economics. Ranking and report-card procedures place their error control directly on the first-stage standard errors; the discrimination report cards of Kline et al. (2024) are a prominent recent example. Meta-analytic and hierarchical pipelines likewise take pairs $(\hat{\theta}_j, s_j)$ as their starting point (Viechtbauer, 2010). When standard errors are not reported, these methods cannot be applied, and when they are improvised, the resulting biases propagate into the downstream analysis.

Existing software does not fill this role. Functions for single-proportion confidence intervals are widely available but do not produce schema-stable panels with joint covariance across indicators. Mixed-model software such as lme4 (Bates et al., 2015) combines measurement and modeling in a single likelihood, which is often appropriate but precludes design-based reporting and the two-stage approaches expected by EB software (Armstrong et al., 2022; Koenker & Gu, 2017). Survey packages presume design weights arising from a probability sample rather than census-type administrative counts. As far as we are aware, no maintained R package treats the sampling uncertainty of site-level summaries as its primary output across both individual-level and aggregate-only inputs.

In this paper we describe sitemix, an R package that converts individual-level records, sufficient counts, or published aggregate tables into a schema-stable table of site-by-year proportions and rates with standard errors, provenance, and, where identified, joint covariance across indicators. The package is confined to measurement: it does not carry out shrinkage, pooling across sites or years, or regression modeling, so its outputs remain valid inputs for whatever analysis follows. It was developed to support accountability analyses of a state pre-kindergarten program and is distributed with a simulated panel that reproduces the reporting structure of such data, but the design is general and applies to any setting in which binary or categorical indicators are summarized over sites, schools, clinics, firms, or neighborhoods. The plan of the paper is as follows. In Section 2 we describe what the package computes, from the supported data-generating scenarios to the output contract and the design principles behind them. In Section 3 we work through the bundled pre-kindergarten panel, ending with a complete EB analysis built on the exported table. Section 4 discusses how

Scenario	Input	Indicator structure	Estimator / covariance
A	records or counts	single binary	binomial $\hat{\theta} = C/n$; reported on a stabilized or raw scale
B	records or joint counts	K overlapping binaries	binomial margins + SUR cross-indicator covariance
C	records or counts	K exclusive categories	multinomial, full simplex covariance
D0	published aggregates	single binary	binomial on published numerator/denominator
D1	published aggregates	K overlapping binaries	marginal binomials; working-independence covariance with recorded dependence provenance

Table 1. The five supported data-generating scenarios. In D1 the cross-indicator covariance is not identified from published marginals: the zeros are labeled assumptions, and `sm_frechet_envelope()` computes formal intervals for the quantities that the marginals do constrain.

the package relates to reporting practice and to neighboring software, together with the evidence supporting the implementation, and Section 5 concludes. The online supplement states every estimator, the output contract, the example panel and the full console session, the audit semantics, and downstream recipes (Appendices A–E).

2. What the package computes

2.1. Five data-generating scenarios. The user names the data-generating scenario explicitly (Table 1); the package does not attempt to infer whether indicators overlap or partition enrollment, since the two cases imply different covariance structures. Scenarios A, B, and C consume individual-level records or sufficient counts, for a single binary indicator, for several overlapping binary indicators, and for mutually exclusive categories respectively. Scenarios D0 and D1 consume published aggregate tables: D0 accepts numerator and denominator pairs for a single indicator, and D1 accepts per-indicator marginal counts for overlapping indicators, the case in which the joint distribution is not identified. We shall return to what can and cannot be recovered in the D1 case at the end of Section 2.2.

2.2. Estimators and their uncertainty. The underlying computations are classical, and we summarize them here; complete statements are given in Appendices A and D and in the package's methodological articles.

We first consider a single binary indicator (Scenario A). For C_{jt} successes among n_{jt} children in site j and year t, the point estimate is always the observed proportion $\hat{\theta}_{jt} = C_{jt}/n_{jt}$, with its plug-in standard error on the probability scale. Since downstream consumers often prefer a variance-stabilized quantity, results are simultaneously reported on a second scale. The default is the arcsine transform $\tilde{\theta}_{jt} = \arcsin\sqrt{\hat{\theta}_{jt}}$, whose first-order standard error $1/(2\sqrt{n_{jt}})$ does not depend on the unknown rate (Anscombe, 1948); logit and identity are available as reporting transforms. At boundary cells ($C_{jt} = 0$ or n_{jt}) the plug-in standard error on the probability scale degenerates to zero, and the package instead records a strictly positive uncertainty surrogate, derived by default from the Wilson score interval (Wilson, 1927) or alternatively from the Agresti–Coull adjusted proportion (Agresti & Coull, 1998), while the point estimate remains the observed proportion; the choice is recorded in provenance. The arcsine-scale standard error, which depends on n_{jt} alone, is unaffected. Cells with small denominators are flagged and retained. When sites are censuses of fixed populations, an optional finite-population correction with simple-random-sampling semantics (Cochran, 1977) propagates through every scale and through whole covariance matrices, and a complete census yields exactly zero sampling uncertainty.

For overlapping indicators (Scenario B), the joint counts identify the cross-indicator sampling covariance $(\hat{p}_{kl} - \hat{p}_k\hat{p}_l)/n$, which has the seemingly-unrelated-regressions structure of Zellner (1962) and is returned per site-year as a $K \times K$ matrix in the `V` list-column. For mutually exclusive categories (Scenario C), the full-simplex multinomial covariance is returned with its analytic rank, the number of observed categories minus one, and with the simplex identity $V\mathbf{1} = 0$ preserved (Agresti, 2013).

When only published tables are available, two further paths apply. D0 handles single indicators from numerator and denominator pairs and reproduces the individual-level results exactly, so nothing is lost by receiving a table in place of records. D1 handles per-indicator marginals for overlapping indicators, for which the joint distribution is not identified. Here the dependence structure is a matter of recorded provenance: the analyst states the sampling relation (`same_units`, `different_units`, or the default `unknown`), equal denominators are never taken to imply a common population, and the returned covariance is an explicitly labeled working-independence diagonal whose zeros are documented as assumptions. For the quantities that the marginals do constrain, `sm_frechet_envelope()` computes the formal Fréchet–Hoeffding intervals (Fréchet, 1951; Hoeffding, 1940; Nelsen, 2006): for each indicator pair, sharp bounds on the joint probability, $\max(0, p_k + p_l - 1) \leq p_{kl} \leq \min(p_k, p_l)$, with the induced covariance and correlation endpoints. Positive-semidefinite projections of the bound corners (Higham, 2002) are additionally available for stress testing when $K > 2$, but they are returned separately and labeled `stress_scenario_not_bound`, since projection can move individual entries outside their own pairwise intervals; they are scenarios under which an analysis may be repeated and should not be read as bounds.

Two further tools round out the estimators. Publisher-suppressed cells are audited before estimation (`sm_suppression_report()`) and preserved as missing-value audit rows by default; a worst-case sensitivity mode exists but requires an explicit acknowledgment flag and writes its results to separate columns that are excluded from ordinary covariance and Fréchet computations. An experimental generalized variance-function smoother (`sm_smooth_variance()`, log-linear or `mgcv`-based; Wood 2017) is opt-in and append-only, because a pre-registered fixed-seed simulation audit did not support promoting smoothing to a default; we return to this in Appendix D.

2.3. **The output contract.** Every engine writes to one canonical S3 class, `sitemix_estimates`: a tibble with 18 schema-stable columns carrying the raw-scale pair, the reporting-scale pair, the denominator, and provenance flags for small cells, boundary cells, suppression, and input mode (the full schema is given in Appendix B). Optional blocks extend the schema without disturbing it: a `V` list-column of covariance carriers, a suppression block, a design block under finite-population corrections, and appended smoothing columns.

The covariance carriers deserve comment, because they are where silent corruption would otherwise enter a long-lived pipeline. Each `V` entry is an `sm_vcov` object that stores, besides the matrix, its scale, method, analytic rank, boundary and correction rules, and the full design block, together with a statement of which scalar column its diagonal must match. The registered print and conversion methods revalidate the carrier before reading, so a corrupted or scale-mismatched covariance produces an error rather than propagating. A three-level audit (`sm_diagnose()`, at object, row, and covariance level) classifies the state of any estimates object under an error, warning, note, ok severity matrix; Appendix D states the semantics.

2.4. **Design principles and architecture.** Four considerations shaped the design. First, the user names the data-generating scenario, as described above. Second, every (site, year, indicator) cell is estimated independently from its own counts, with no pooling, so each output row depends only on that cell and two analysts working from the same extract obtain identical tables. Third, measurement is kept separate from

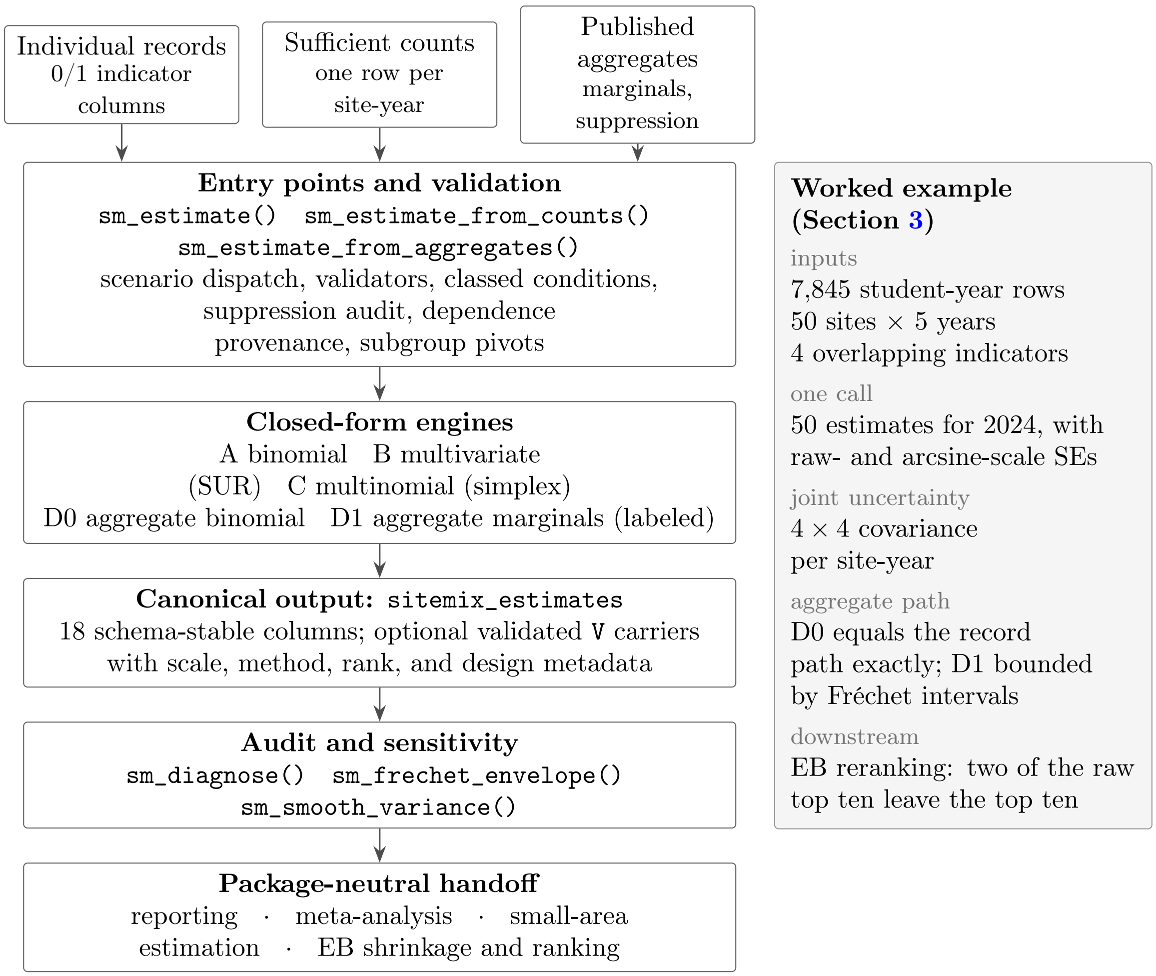

FIGURE 1. Architecture of sitemix v0.3.0, with the worked example of Section 3 alongside. Individual-level and aggregate-only inputs converge on one dispatcher; five closed-form engines write to a single canonical schema; audit tools examine the result before the package-neutral handoff.

modeling: shrinkage, pooling, and regression belong downstream, and the package is neutral about which downstream consumer follows. Fourth, claims that the data cannot support fail closed. Unidentified covariance is labeled as an assumption instead of being filled in, suppressed cells yield audit rows instead of fabricated values, and all user-facing errors are classed conditions on which pipelines can branch.

sitemix is implemented in pure R, with no compiled code and five lightweight dependencies (rlang, tibble, vctrs, cli, Matrix), so it can be installed in restricted agency computing environments. Figure 1 summarizes the architecture: two families of inputs are handled by a single dispatcher, five closed-form engines write to one canonical

Function	Purpose
`sm_estimate()`	Main dispatcher: site-year estimates from records, counts, or aggregates (Scenarios A/B/C/D0/D1)
`sm_estimate_from_counts()`	Wrapper for sufficient-count input (one row per site-year)
`sm_estimate_from_aggregates()`	Wrapper for published aggregate rows with D0/D1 dispatch and dependence provenance
`sm_diagnose()`	Object-, row-, or covariance-level audit with an error/warning/note/ok severity matrix
`sm_suppression_report()`	Publisher-side suppression audit before estimation
`sm_vcov()`	Constructor/validator for covariance carriers with scale, method, rank, and design metadata
`sm_frechet_envelope()`	Formal Fréchet–Hoeffding pairwise intervals plus labeled projection stress scenarios
`sm_smooth_variance()`	Experimental, append-only generalized variance-function smoothing
`sm_pivot_subgroups_to_sites()`	Reshape subgroup rows to subgroup-as-site aggregates
`sm_pivot_subgroups_to_indicators()`	Reshape subgroup rows to subgroup-as-indicator aggregates
`prek_sim` (data)	Simulated 50-site, 2021–2025 pre-K panel with four indicators

TABLE 2. Exported functions and data in sitemix v0.3.0.

schema, and the audit tools examine the result before it leaves the package. Table 2 lists the exported surface, which consists of 10 functions and the bundled data set.

3. The package in use

We illustrate the package using `prek_sim`, a simulated pre-kindergarten site panel distributed with it. The panel is generated from design constants recorded in the package, reads no external file, and describes no real children, sites, or program; its structure is chosen to exhibit the features the estimators address, and Appendix C describes the generative design. The data comprise 7,845 student-year rows covering 50 sites over 2021–2025, with four overlapping means-tested binary indicators (`frpm`, `snap`, `wic`, `tanf`), and site-year cells ranging from 6 to 134 children. All output below was produced with R 4.6.0 and sitemix 0.3.0; every computation is closed-form and deterministic. The complete session is reproduced in Appendix C and in the manuscript repository.

3.1. A single indicator, and what the intervals change.

A single call converts individual records into estimates with standard errors:

```
library(sitemix)
data(prek_sim, package = "sitemix")
```

```
a24 <- subset(prek_sim, year == 2024)

est <- sm_estimate(a24, family = "binomial", indicator = "frpm")
head(est, 5)
```

```
sitemix_estimates: 5 rows x 18 columns | family=binomial | role=summary_uncertainty
groups=5 sites=5 years=1 indicators=1 V=FALSE K=FALSE
# A tibble: 5 x 18
  site_id  year indicator theta_raw theta_hat se_raw    se     n n_eff
1 S001     2024 frpm          0.111     0.340  0.105 0.167     9     9
2 S002     2024 frpm          0.8       1.11   0.126 0.158    10    10
3 S003     2024 frpm          0.5       0.785  0.177 0.177     8     8
4 S004     2024 frpm          0.429     0.714  0.132 0.134    14    14
5 S005     2024 frpm          0.875     1.21   0.117 0.177     8     8
# ... with 9 more columns: estimate_scale, transform, var_method,
#   flag_small_n, flag_zero_cell, input_mode, flag_suppressed, framing,
#   flag_below_accountability
```

Each row carries the observed proportion (**theta_raw**) with its probability-scale standard error, the arcsine-scale pair (**theta_hat** and **se** $= 1/(2\sqrt{n})$), and provenance flags. Figure 2 shows what attaching the intervals changes. The median 2024 enrollment is 24 children, and the 95% interval of the highest-ranked site overlaps the intervals of 23 of the other 49 sites, so the apparent ordering near the top of panel (a) is largely unsupported once sampling uncertainty is displayed in panel (b).

Figure 3 shows the two reported scales for all 250 site-year cells. On the probability scale the standard error depends on the underlying rate as well as on enrollment, and at the two boundary cells (a site-year with no FRPM children) the plug-in value would be zero; the package reports the Wilson surrogate there instead, and flags the rows. On the arcsine scale the standard error is $1/(2\sqrt{n})$ at every cell, boundary cells included, which is the property that makes this scale convenient for downstream methods that assume known sampling variances. Before the table is passed on, a one-line audit summarizes its state:

```
> sm_diagnose(est, level = "summary")
sitemix_diagnostics_summary: 50 cells | 50 groups | 1 indicators |
scalar finite=TRUE SE positive=TRUE scales consistent=TRUE V valid=NA
sensitivity role=none V stale=FALSE severity=note
```

3.2. **Joint uncertainty across indicators.** Because FRPM, SNAP, WIC, and TANF overlap within children, any analysis that combines them requires their joint sampling covariance, which Scenario B computes in closed form:

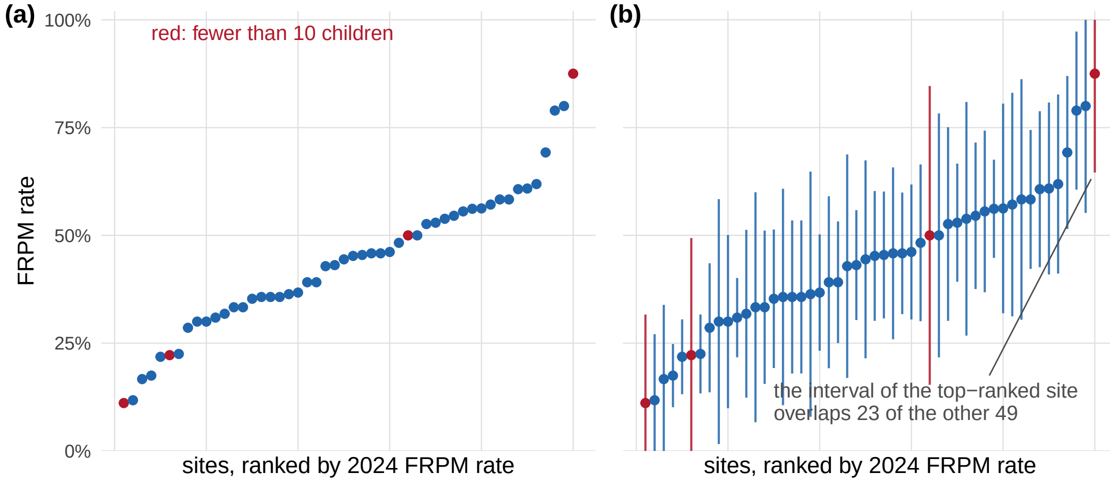

FIGURE 2. Site-level FRPM rates for 2024 in ranked order, (a) as they would appear in a published league table and (b) with the 95% intervals that sitemix attaches to every estimate. Sites with fewer than 10 children are shown in red.

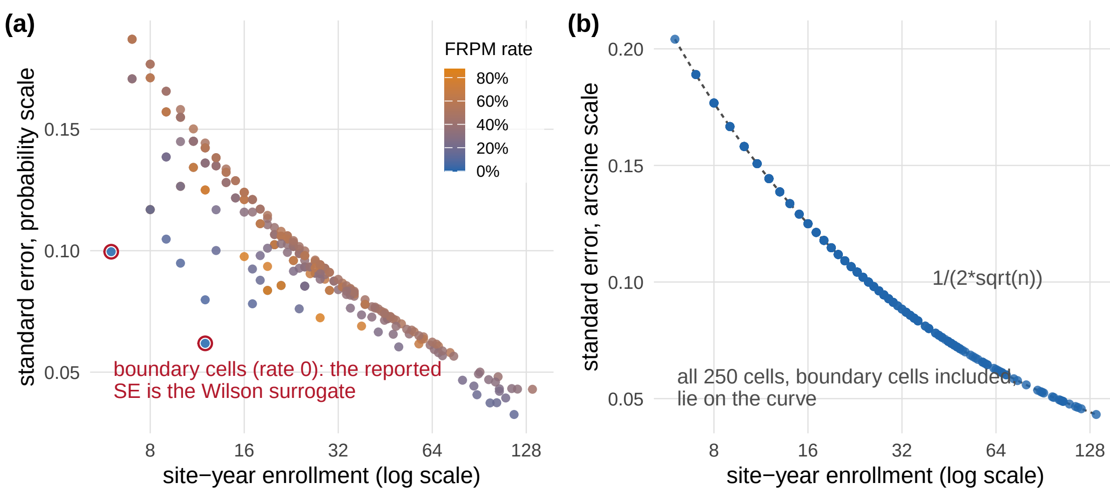

FIGURE 3. Standard errors for all 250 site-year cells of the FRPM indicator, against enrollment. (a) On the probability scale the standard error varies with the underlying rate at every n, and the two boundary cells (red circles) carry Wilson surrogates in place of a degenerate plug-in value of zero. (b) On the arcsine scale every cell lies on the first-order curve $1/(2\sqrt{n})$.

```
est_mv <- sm_estimate(a24, family = "multivariate",
                      indicators = c("frpm", "snap", "wic", "tanf"),
```

```
                         vjt = TRUE)
est_mv$V[[which(est_mv$site_id == "S028")[1]]]
```

```
sm_vcov[4x4] site_id=S028 year=2024 family=multivariate
             vcov_method=sur vcov_scale=raw
          frpm     snap       wic       tanf
frpm  0.008193 0.003300 0.0015930 -0.0005690
snap  0.003300 0.008705 0.0021620  0.0010240
wic   0.001593 0.002162 0.0091030  0.0003414
tanf -0.000569 0.001024 0.0003414  0.0027310
```

The positive `frpm`–`snap` covariance at this 26-child site reflects the overlap of the two eligibility populations; a working-independence assumption would discard exactly this information. Mutually exclusive categories are handled in the same way. Deriving a depth-of-participation variable (none / one–two / three-plus programs) and calling `sm_estimate(family = "multinomial")` returns per-category shares with the full simplex covariance, whose recorded rank equals the number of observed categories minus one and whose rows sum to zero by construction.

3.3. **Published tables: what survives aggregation.** We next turn to aggregate-only data, and Figure 4 carries the two findings. Aggregating the panel to published-style count columns and calling `sm_estimate_from_aggregates()` (Scenario D0) reproduces the individual-level results exactly, column for column, across all 250 site-year cells (panel a); the equivalence is convenient when agencies can share only tables. When a publisher releases only per-indicator marginals, Scenario D1 estimates each margin and records what it cannot know:

```
est_d1 <- sm_estimate_from_aggregates(
  d1_long,                          # site_id, year, indicator, c_jt, n_jt
  family             = "multivariate",
  indicator_col      = "indicator",
  sampling_relation = "same_units",
  vjt                = TRUE
)
env <- sm_frechet_envelope(est_d1, population_regime = "d1a")
```

```
(classed warning: sitemix_warning_working_independence_default)
attributes: sampling_relation = same_units | d1_regime = D1a |
            denominator_pattern = common
> est_d1$V[[1]]
sm_vcov[4x4] ... vcov_method=working_independence vcov_scale=arcsine_delta
(diagonal matrix; off-diagonal zeros are labeled assumptions)
```

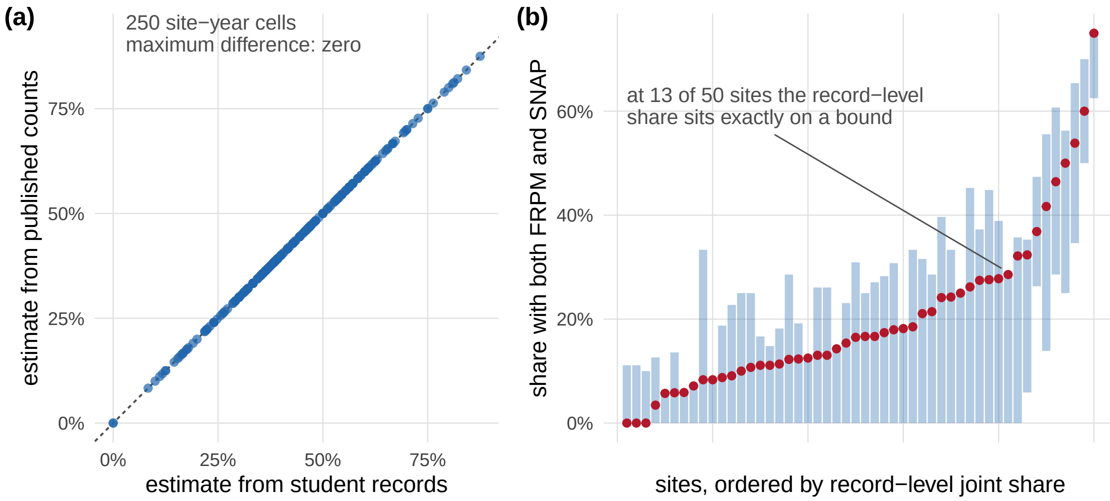

FIGURE 4. What published tables preserve and what they only constrain. (a) For a single indicator, estimates computed from published counts (D0) equal the estimates computed from student records at all 250 site-year cells. (b) For the joint FRPM–SNAP share, published marginals identify only the Fréchet interval (bars); the record-level value (dots) lies inside the interval at every site.

```
> head(env$raw_pairwise_intervals)   # interval_scope: formal_raw_pairwise_interval
 site_id  p_1  p_2 n_common joint_probability_lower joint_probability_upper
    S001 0.111 0.222        9                     0.0                   0.111
    S002 0.800 0.700       10                     0.5                   0.700
```

The declared sampling relation has operational consequences: only `same_units` data qualify for the formal Fréchet intervals, whereas the default `unknown` restricts the analysis to what weaker provenance supports. Panel (b) of Figure 4 shows the result for all 50 sites. From the marginals alone, the joint FRPM–SNAP share is confined to the formal interval; the value computed from the individual-level records, which an aggregate-only analyst never observes, lies within the interval at every site, and at 13 of the 50 sites it sits exactly on a bound.

3.4. **A downstream analysis on the exported table.** Finally, we illustrate a downstream analysis. Since the output is package-neutral, a complete normal–normal EB step of the kind described by Walters (2024) and Kline et al. (2024) requires only six lines of base R applied to the exported table:

```
d <- as.data.frame(est)                          # transport, not validation
d <- d[is.finite(d$theta_hat) & d$se > 0, ]
```

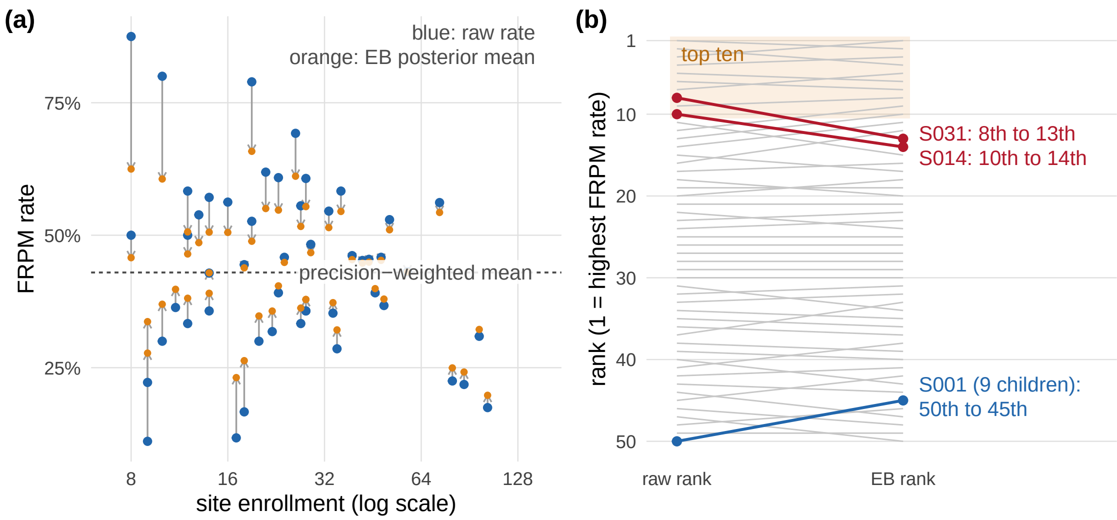

FIGURE 5. The downstream EB step computed in six lines of base R from the exported table. (a) Posterior means pull each site's raw 2024 FRPM rate toward the precision-weighted mean, and the pull decreases with enrollment. (b) The same movement in rank terms: two sites leave the top ten (red), and the lowest-ranked site rises five places (blue).

```
tau2 <- max(0, var(d$theta_hat) - mean(d$se^2))   # method of moments
mu   <- weighted.mean(d$theta_hat, 1 / (tau2 + d$se^2))
lam  <- tau2 / (tau2 + d$se^2)                     # shrinkage weight
d$theta_eb <- mu + lam * (d$theta_hat - mu)
```

```
tau^2 = 0.0206 | mu (arcsine) = 0.715 | shrinkage weight: min 0.40 / max 0.89
Spearman correlation of raw vs EB ranks: 0.988
Sites in the raw top 10 that stay in the EB top 10: 8 of 10
largest rank movements:
 site_id  n p_raw p_eb rank_raw rank_eb
    S001  9  0.11 0.28       50      45
    S031 12  0.58 0.51        8      13
    S012 12  0.33 0.38       37      33
```

Figure 5 displays the shrinkage and the resulting ranks. The results illustrate the practical stakes of the first stage. Although the rank correlation between the raw and EB orderings is high (0.988), two of the ten highest-ranked sites leave the top ten once uncertainty is taken into account: a twelve-child site falls from rank 8 to rank 13, a ten-child site from rank 10 to rank 14. At the other end, the lowest-ranked site, with nine children, rises from rank 50 to rank 45. These reorderings cannot be anticipated from the point estimates alone.

4. Positioning and impact

The most direct impact of sitemix is on reporting practice. The package makes it no harder to report a site-level rate with its sampling uncertainty than without, which is the convention that survey data users already take for granted (U.S. Census Bureau, 2020). Choices that are usually made informally, concerning transformation, boundary handling, denominators, suppression, and dependence, become named arguments with tested defaults, recorded in provenance columns, so a methods section can cite an estimator instead of describing a spreadsheet. Because every cell is estimated independently by closed-form rules, two analysts working from the same extract obtain identical tables.

A second consequence is a wider reach for uncertainty-aware analysis. The aggregate paths accept the count tables that agencies actually publish, so rigorous uncertainty quantification becomes available to analysts who never hold individual-level data. The D0 equivalence guarantees that nothing is lost for single indicators, and for joint quantities the Fréchet machinery converts a limitation of aggregate data that is usually invisible into an explicit, computable interval (Figure 4). In our own work, the workflow that supports internal analyses of a state pre-kindergarten panel applies unchanged to public state education tables such as chronic-absenteeism counts.

The package also supplies a measurement layer for comparison methods that are moving from econometrics into routine program accountability (Kline et al., 2024; Walters, 2024). These methods consume exactly what sitemix produces, and the demonstration in Section 3 suggests that respecting uncertainty can reorder a league table in ways that matter for any consequence attached to it. Because the output contract is package-neutral, the same table can be passed to REBayes (Koenker & Gu, 2017), ebci (Armstrong et al., 2022), meta-analytic tools (Viechtbauer, 2010), or hierarchical models (Bates et al., 2015) without further preparation.

Several features are directed at institutional settings in which tables are maintained and reused over long periods: the three-level diagnostic audit, classed errors and warnings, fail-closed handling of suppressed and unidentified quantities, and covariance carriers that revalidate themselves on access. The documentation comprises seventeen long-form articles, an applied track (A1–A9) and a methodological track (M1–M8) with complete derivations and the package-neutral output contract, and we have found it suitable as teaching material for graduate research assistants.

Finally, we summarize the evidence supporting the implementation. Version 0.3.0 is verified by a test suite of 53 files with 6,899 passing expectations, zero failures, and

91.4% line coverage; it passes `R CMD check` with no errors, warnings, or notes, and the repository's continuous-integration matrix, which spans three operating systems and R versions back to 4.1, passes on the published commit. Before the v0.2.0 release the candidate underwent an independent external code review comprising 24 findings, all resolved; the single runtime defect it surfaced, a dispatch guard in the mapped aggregate path, was corrected and locked with a public regression test. Run time is modest at realistic scales: the bundled benchmarks process 1,000-row workloads with full joint covariance in about two seconds on a laptop.

5. Concluding remarks

We have described an R package that attaches sampling uncertainty to site- and group-level proportions and rates computed from administrative data, working from individual records, sufficient counts, or published aggregates within a single canonical schema. The scope is narrow: the package estimates proportions and rates only, does not carry out shrinkage or pooling, and reports identification limits instead of resolving them by assumption. In our view this restraint is what makes the output a dependable input to downstream reporting, meta-analysis, small-area estimation, and empirical Bayes methods. Planned work includes submission to CRAN and, as demand warrants, extension of the estimand family beyond proportions and rates, which we have deferred until the same uncertainty guarantees can be provided there. More generally, the same treatment applies wherever administrative rates are reported, and it makes their margins of error as available as they already are for survey estimates.

Software and reproducibility. sitemix is open-source software released under the MIT license (project home page https://github.com/joonho112/sitemix; documentation at https://joonho112.github.io/sitemix/), requiring R ($\geq$ 4.1.0) and containing no compiled code. Every number, table, and figure in this article is regenerated deterministically from the installed package by the scripts `scripts/analysis.R` and `scripts/figures.R` in the manuscript repository, and `scripts/reproduce.R` asserts the reported values against the package's output. An archived release will be deposited at Zenodo on journal submission.

Data availability. The example data set `prek_sim` is fully simulated and is distributed with the package together with the builder script that regenerates it from its design constants (Appendix C). No administrative or person-level records are used or distributed.

Competing interests. The authors declare that they have no competing interests.

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Authors' contributions. JL conceived and developed the software, performed the analysis, and drafted the manuscript. AH contributed to the conceptualization, methodology, and validation, provided resources and supervision, and revised the manuscript. Both authors read and approved the final manuscript.

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Online Supplement

sitemix: An R Package for Site- and Group-Level Proportions, Rates, and Sampling Uncertainty from Individual and Aggregate Administrative Data

JoonHo Lee   Alison Hooper

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Appendix A. Estimators by Scenario

This appendix states what sitemix v0.3.0 computes for each of the five supported data-generating scenarios. The package's methodological articles (M1–M8) give complete derivations; here we fix notation and record the estimators and their contracts.

A.1. **Setting.** For site j, year t, and indicator k, let θ_{jt} denote the underlying rate and $(\hat{\theta}_{jt}, s_{jt})$ the reported estimate and standard error. Every (site, year, indicator) cell is estimated independently from its own counts; no pooling occurs. The package's role attribute, `summary_uncertainty`, marks its outputs as measured summaries with sampling uncertainty, that is, inputs to rather than outputs of any downstream model.

A.2. **Scenario A: single binary indicator (binomial).** With C_{jt} successes among n_{jt} children, $C_{jt} \sim \text{Binomial}(n_{jt}, \theta_{jt})$:

$$\hat{\theta}^{\text{raw}}_{jt} = \frac{C_{jt}}{n_{jt}}, \qquad s^{\text{raw}}_{jt} = \sqrt{\frac{\hat{\theta}^{\text{raw}}_{jt}(1-\hat{\theta}^{\text{raw}}_{jt})}{n_{jt}}}. \tag{A.1}$$

The point estimate is *always* the observed proportion; no boundary rule or transform changes it. Alongside the raw pair, we report results on a second scale selected by `vst`: the default arcsine transform

$$\hat{\theta}_{jt} = \arcsin\sqrt{\hat{\theta}^{\mathrm{raw}}_{jt}}, \qquad s_{jt} = \frac{1}{2\sqrt{n_{jt}}} \tag{A.2}$$

is the only variance-stabilizing option, and $1/(4n)$ is documented as a first-order (delta-method) working variance rather than an exact finite-sample variance (Anscombe, 1948); logit (with delta-method standard error, interior cells only) and identity are *reporting transforms* and are not variance-stabilizing. An optional Anscombe-type small-sample adjustment, available for the arcsine scale only, uses the shifted proportion $(C_{jt} + 3/8)/(n_{jt} + 3/4)$.

Boundary cells. At $C_{jt} \in \{0, n_{jt}\}$ the plug-in standard error on the probability scale degenerates to zero. We substitute a strictly positive 'uncertainty surrogate', derived from the Wilson score interval by default (Wilson, 1927) or from the Agresti–Coull adjusted proportion $(C_{jt} + 2)/(n_{jt} + 4)$ (Agresti & Coull, 1998), while `theta_raw` and `theta_hat` keep the observed value; the arcsine-scale standard error in Equation (A.2), which depends on n_{jt} alone, does not degenerate and is reported unchanged. The surrogate that fired is recorded in provenance (`wilson_boundary_surrogate` / `agresti_coull_boundary_surrogate`) and flagged in `flag_zero_cell`. Cells with n_{jt} below `min_n` (default 10) or the reporting threshold `accountability_n` (default 30) are flagged, never dropped.

Finite populations. The optional `fpc` argument is the fixed population size N_{jt} under simple random sampling without replacement (Cochran, 1977): variances are multiplied by $(N_{jt} - n_{jt})/(N_{jt} - 1)$-type factors, the correction propagates through every reported scale and through whole covariance matrices, and a complete census ($n_{jt} = N_{jt}$) yields exact zero sampling uncertainty, which the diagnostics report as a note rather than an error. Outputs gain a design block (population size, sampling fraction, multipliers, design, variance rule) so the correction is auditable. An optional small-sample bias correction (`binomial_bc`) likewise applies to scalar standard errors and whole matrices coherently.

A.3. **Scenario B: overlapping binary indicators (multivariate Bernoulli).** When K indicators overlap within children, the joint counts identify the cross-indicator sampling covariance. For indicators k and l with joint success proportion $\hat{p}_{kl}$,

$$\widehat{\mathrm{Cov}}\big(\hat{\theta}^{\mathrm{raw}}_k, \hat{\theta}^{\mathrm{raw}}_l\big) = \frac{\hat{p}_{kl} - \hat{p}_k\,\hat{p}_l}{n_{jt}}, \tag{A.3}$$

the seemingly-unrelated-regressions structure of Zellner (1962). We return the $K \times K$ matrix per site-year in the `V` list-column on the *raw* probability scale (`vcov_scale = "raw"`), with the diagonal contract `row_se_raw_squared`: $\mathrm{diag}(V)$ matches the squared raw standard errors rather than the reported-scale `se`^{2}, unless the scales coincide. In the worked example of Appendix C, site S001 in 2024 has nine children, of whom one receives FRPM and two receive SNAP, with no child receiving both: $\hat{p}_{\mathrm{frpm}} = 1/9$, $\hat{p}_{\mathrm{snap}} = 2/9$, and $\hat{p}_{kl} = 0$, so Equation (A.3) gives $(0 - 2/81)/9 = -0.00274$, the value visible in the printed matrix. Joint feasibility of supplied sufficient counts is checked exactly for $K \leq 3$; for $K \geq 4$, count-only inputs whose joint feasibility cannot be verified fail closed (individual-level records, whose joint counts are directly observed, are unaffected).

A.4. **Scenario C: mutually exclusive categories (multinomial).** For a partition into K categories with counts $(C_1, \ldots, C_K)$ and $\hat{\boldsymbol{p}} = (C_1, \ldots, C_K)/n$, the full-simplex covariance

$$\widehat{\mathrm{Cov}}(\hat{\boldsymbol{p}}) = \frac{\mathrm{diag}(\hat{\boldsymbol{p}}) - \hat{\boldsymbol{p}}\hat{\boldsymbol{p}}^{\top}}{n} \tag{A.4}$$

is returned (Agresti, 2013); we preserve the simplex identity $V\mathbf{1} = 0$ and record the *analytic* rank, the number of categories with positive support minus one, so a site observing only two of three categories carries rank 1 rather than a numerical artifact. The session in Appendix C shows such a case: S001 observes two of the three derived support tiers, and its carrier records `matrix_rank = 1` with `positive_support = 2`.

A.5. **Scenarios D0 and D1: published aggregates.** D0 accepts published numerator/denominator pairs for a single indicator and routes them through the Scenario-A engine; Appendix C verifies that D0 reproduces the individual-record results exactly, column for column, across all 250 site-year cells.

D1 accepts per-indicator *marginal* counts for overlapping indicators. Its diagonal follows from the marginals; the off-diagonal covariance depends on joint proportions that the publisher did not release, and we therefore return an explicitly labeled working-independence diagonal (`vcov_method = "working_independence"`), whose zeros are assumptions imposed for want of the joint counts. The dependence structure is a matter of *recorded provenance*: the analyst declares `sampling_relation` as `"same_units"` (the indicators were measured on one common set of units), `"different_units"`, or the default `"unknown"`, mapping to regimes D1a, D1b, and unknown respectively. Equal denominators are recorded separately (`denominator_pattern`) and never taken to imply a common population. The declaration gates everything downstream: only `same_units` data qualify for the formal Fréchet analysis of Appendix D, and a classed warning marks every working-independence default. We also require one complete, identically ordered indicator set across site-year groups; heterogeneous sets fail closed with a classed error.

Appendix B. Output Contract

Every engine writes to one canonical S3 class, `sitemix_estimates`: a tibble with the 18 schema-stable columns of Table B.1, plus documented optional blocks. We lock the schema with snapshot and regression tests, and the object records its role (`sitemix_role = "summary_uncertainty"`), family, and, for aggregate inputs, the dependence provenance attributes (`sampling_relation`, `denominator_pattern`, `d1_regime`).

Column	Type	Meaning
`site_id`	chr	Site identifier (key)
`year`	int	Year (key)
`indicator`	chr	Indicator name (key)
`theta_raw`	dbl	Observed proportion C/n
`theta_hat`	dbl	Point estimate on the reporting scale (`estimate_scale`)
`se_raw`	dbl	Standard error on the probability scale
`se`	dbl	Standard error on the reporting scale
`n`	int	Denominator (enrollment)
`n_eff`	dbl	Effective denominator (transform provenance)
`estimate_scale`	chr	`arcsine`, `logit`, or `identity`
`transform`	chr	Transform actually applied
`var_method`	chr	Variance estimator / surrogate provenance
`flag_small_n`	lgl	n below `min_n` (default 10)
`flag_zero_cell`	lgl	Boundary cell; an uncertainty surrogate fired
`input_mode`	chr	Records, counts, or aggregates
`flag_suppressed`	lgl	Cell affected by publisher suppression
`framing`	chr	Subgroup framing (pivot paths), else `NA`
`flag_below_accountability`	lgl	n below `accountability_n` (default 30)

TABLE B.1. The 18 schema-stable columns of `sitemix_estimates`. Raw- and reporting-scale quantities always coexist; quality issues are flagged and never dropped.

Optional blocks. With `vjt = TRUE` the tibble gains a `V` list-column of covariance carriers (and `K`, the indicator count). When aggregate inputs contain suppressed cells, a suppression block is added (`estimate_status` distinguishing identified, suppressed-missing, and sensitivity rows, plus `sensitivity_*` columns; Appendix D). When `fpc` is supplied, a design block records population size, sampling fraction, variance and SE multipliers, design, and variance rule. Smoothing appends `se_smoothed` (or `se_raw_smoothed`) and `var_method_smoothed` without touching canonical columns.

The covariance carrier. Each `V` entry is an `sm_vcov` object that stores, besides the matrix itself, its scale (`raw`, `arcsine_delta`, `logit_delta`), method (`sur`, `multinomial`, `working_independence`), analytic rank and positive support, boundary and correction rules, any positive-semidefiniteness repair, the denominators, and the full design block (24 metadata fields in total), together with a `diag_contract` stating exactly which scalar column the diagonal must match. We enforce validation with scale-aware tolerances (separate thresholds for symmetry, positive semidefiniteness, the simplex identity, and numerical rank, each relative to the matrix scale), and the registered `print()`/`as.matrix()` methods revalidate before reading, so a corrupted matrix fails loudly rather than propagating.

The neutral handoff. The documented consumption pattern (articles A8 and M8) is: audit with `sm_diagnose()`; select IDs with `theta_hat`/`se` *or* `theta_raw`/`se_raw` together with provenance columns; pair estimates only with a covariance of compatible scale; evaluate linear contrasts as $a^\top V a$ without inverting (so singular multinomial and exact-census matrices remain consumable); and treat `as.data.frame()` as a transport operation that performs no validation. Inverse-variance weighting is restricted to identified rows with finite positive standard errors; census-zero, suppressed, and sensitivity rows are retained as labeled states rather than receiving infinite weight.

Appendix C. The Example Panel and the Complete Session

C.1. **The simulated panel.** `prek_sim` is generated by a builder script shipped with the package (`inst/scripts/build-prek-sim.R`) that reads no external file: every quantity it uses is a design constant declared in the script, so any user can regenerate the shipped artifacts from a plain R installation, and the package's provenance audit verifies the rebuild byte for byte. The design targets are round numbers chosen so that the panel exhibits the features the estimators address; they are not estimates of any particular program's caseload.

The generative design has three components. Site sizes come from three strata (10 small, 20 medium, and 20 large sites), with each site's median cell size drawn from a stratum-specific lognormal distribution and its five yearly cells varying lognormally around that median. Each site carries a vector of correlated random effects on the logit scale, one per indicator, so a site that serves a high-need population tends to do so on every indicator at once. Within a site-year cell, the four binary indicators are drawn per child from a Gaussian copula thresholded at the cell's indicator probabilities, which produces the within-child overlap (a child receiving both FRPM and SNAP) that Scenario B measures. The marginal rates, between-site spreads, and pairwise indicator correlations of the realized panel are calibrated to the declared round targets by solving for the latent parameters; the calibration residuals are recorded in the panel's `build_info` attribute and are below 10^{-3} on every target.

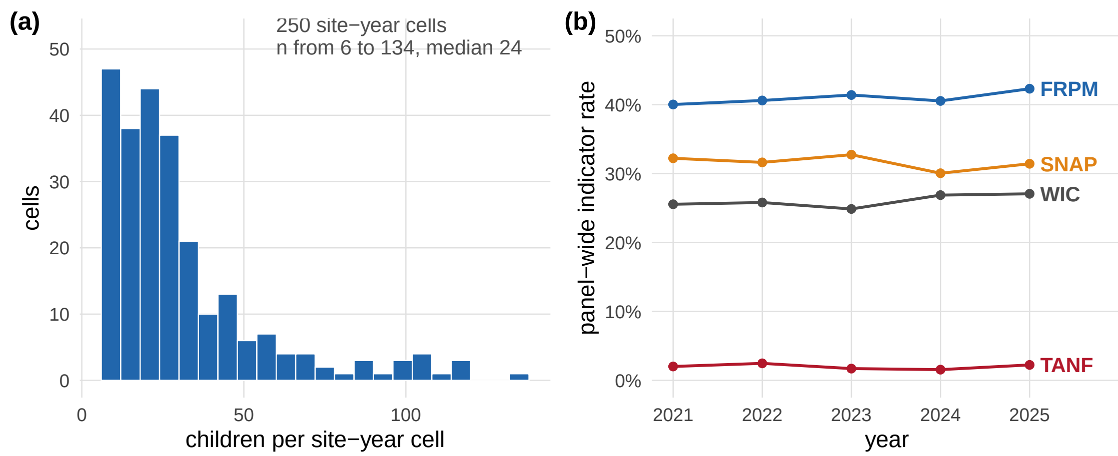

FIGURE C.1. Structure of the simulated panel. (a) The 250 site-year cell sizes range from 6 to 134 children with median 24; 21 cells fall below 10. (b) Panel-wide indicator rates are stable across the five years, with TANF rare by design (157 of the 250 cells contain no TANF child, the sparse-cell regime that the boundary and suppression machinery is built for).

Figure C.1 summarizes the realized structure: 7,845 student-year rows across 50 sites and five years, cell sizes from 6 to 134 (median 24), and one rare indicator (TANF, at two percent overall) that leaves 157 of the 250 cells with a zero count. Two cells of the FRPM indicator are boundary cells with no FRPM child, which is what exercises the Wilson surrogate in Section 3 of the main text.

C.2. **The console session.** The remainder of this appendix reproduces the console transcript behind Section 3 of the main text (R 4.6.0, sitemix 0.3.0). The script is `scripts/analysis.R` in the manuscript repository. All computations are closed-form and deterministic; the transcript is verbatim except that non-ASCII console glyphs are transliterated and tibble type-annotation rows are trimmed. Informational cli messages print to standard error and are shown where they appeared on screen.

C.3. **Scenario A and diagnostics.** The session opens with the Scenario-A call of the main text and its one-line audit.

```
library(sitemix)
data(prek_sim, package = "sitemix")
a24 <- subset(prek_sim, year == 2024)

est <- sm_estimate(a24, family = "binomial", indicator = "frpm")
head(est, 5)
```

```
sitemix_estimates: 5 rows x 18 columns | family=binomial | role=summary_uncertainty
groups=5 sites=5 years=1 indicators=1 V=FALSE K=FALSE
# A tibble: 5 x 18
  site_id  year indicator theta_raw theta_hat se_raw    se     n n_eff
1 S001     2024 frpm          0.111     0.340  0.105 0.167     9     9
2 S002     2024 frpm          0.8       1.11   0.126 0.158    10    10
3 S003     2024 frpm          0.5       0.785  0.177 0.177     8     8
4 S004     2024 frpm          0.429     0.714  0.132 0.134    14    14
5 S005     2024 frpm          0.875     1.21   0.117 0.177     8     8
# ... with 9 more columns: estimate_scale, transform, var_method,
#   flag_small_n, flag_zero_cell, input_mode, flag_suppressed, framing,
#   flag_below_accountability
```

```
sm_diagnose(est, level = "summary")
```

```
sitemix_diagnostics_summary: 50 cells | 50 groups | 1 indicators |
scalar finite=TRUE SE positive=TRUE scales consistent=TRUE V valid=NA
sensitivity role=none V stale=FALSE severity=note
# A tibble: 1 x 40
  family   sitemix_role        n_cells n_groups n_sites n_years n_indicators
1 binomial summary_uncertainty      50       50      50       1            1
# ... with 33 more variables: n_flag_small_n, n_flag_zero_cell,
#   n_flag_both, n_flag_suppressed, n_flag_below_accountability,
#   n_identified, n_suppressed_missing, n_suppression_sensitivity,
#   n_zero_uncertainty_census, min_n, median_n, max_n, estimate_scale,
#   v_present, k_present, n_psd_repair_fired, scalar_uncertainty_finite, ...
```

C.4. **Scenario B: joint covariance.** Joint covariance for the four overlapping indicators, shown for the nine-child site S001.

```
est_mv <- sm_estimate(a24, family = "multivariate",
                      indicators = c("frpm", "snap", "wic", "tanf"),
                      vjt = TRUE)
head(est_mv, 4)
```

```
sitemix_estimates: 4 rows x 20 columns | family=multivariate | role=summary_uncertainty
groups=1 sites=1 years=1 indicators=4 V=TRUE K=TRUE
# A tibble: 4 x 20
  site_id  year indicator theta_raw theta_hat se_raw    se     n n_eff
1 S001     2024 frpm          0.111     0.340 0.105  0.167     9     9
2 S001     2024 snap          0.222     0.491 0.139  0.167     9     9
3 S001     2024 wic           0.111     0.340 0.105  0.167     9     9
4 S001     2024 tanf          0         0     0.0763 0.167     9     9
# ... with 11 more columns: ..., V <list>, K <int>
```

```
est_mv$V[[1]]
```

```
sm_vcov[4x4] site_id=S001 year=2024 family=multivariate
             vcov_method=sur vcov_scale=raw
          frpm      snap       wic     tanf
frpm  0.010970 -0.002743 -0.001372 0.000000
snap -0.002743  0.019200  0.009602 0.000000
wic  -0.001372  0.009602  0.010970 0.000000
tanf  0.000000  0.000000  0.000000 0.005824
matrix_rank=4 psd_repair=none estimate_scale=arcsine
matrix_boundary_rule=none indicators=frpm,snap,wic,tanf
```

The S001 matrix is the worked example of Appendix A: one of the nine children receives FRPM and two receive SNAP, no child receives both, and the printed `frpm`–`snap` entry equals $(0 - (1/9)(2/9))/9 = -0.00274$. The `tanf` row is a boundary case (no TANF child), so its raw standard error is the Wilson surrogate, recorded in `scalar_correction_rule`, and its covariances with the other indicators are zero.

C.5. **Scenario C: multinomial on derived exclusive categories.** A depth-of-participation partition and its simplex covariance.

```
k_ind <- a24$frpm + a24$snap + a24$wic + a24$tanf
a24$support_tier <- ifelse(k_ind == 0, "none",
                           ifelse(k_ind <= 2, "one_two", "three_plus"))
est_c <- sm_estimate(a24, family = "multinomial",
                     indicator = "support_tier", vjt = TRUE)
```

```
  site_id year  indicator theta_raw se_raw  n
```

```
1    S001 2024       none     0.667 0.1571  9
2    S001 2024    one_two     0.333 0.1571  9
3    S001 2024 three_plus     0.000 0.0763  9
4    S002 2024       none     0.100 0.0949 10
5    S002 2024    one_two     0.600 0.1549 10
6    S002 2024 three_plus     0.300 0.1449 10

Simplex covariance for S001: matrix_rank = 1 | positive_support = 2 |
V %*% 1 ~ 0: TRUE
```

C.6. **Scenario D0: aggregate counts reproduce microdata.** The equivalence check for published counts, over all 250 site-year cells.

```
D0 equals student-row Scenario A on shared columns: TRUE
```

C.7. **Scenario D1 and the Fréchet analysis.** The marginal-only path: the labeled working-independence covariance, then the formal intervals.

```
est_d1 <- sm_estimate_from_aggregates(
  d1_long,                       # site_id, year, indicator, c_jt, n_jt
  family            = "multivariate",
  indicator_col     = "indicator",
  sampling_relation = "same_units",
  vjt               = TRUE
)
```

```
(classed warning: sitemix_warning_working_independence_default)
attributes: sampling_relation = same_units | d1_regime = D1a |
            denominator_pattern = common
```

```
est_d1$V[[1]]
```

```
sm_vcov[4x4] site_id=S001 year=2024 family=multivariate
             vcov_method=working_independence vcov_scale=arcsine_delta
        frpm    snap    tanf     wic
frpm 0.02778 0.00000 0.00000 0.00000
snap 0.00000 0.02778 0.00000 0.00000
tanf 0.00000 0.00000 0.02778 0.00000
wic  0.00000 0.00000 0.00000 0.02778
matrix_rank=4 psd_repair=none estimate_scale=arcsine
matrix_boundary_rule=none indicators=frpm,snap,tanf,wic
```

```
env <- sm_frechet_envelope(est_d1, population_regime = "d1a")
```

```
head(env$raw_pairwise_intervals)   # frpm-snap pairs shown
```

```
interval_scope values: formal_raw_pairwise_interval
 site_id   p_1   p_2 n_common joint_probability_lower joint_probability_upper
    S001 0.111 0.222        9                     0.0                   0.111
    S002 0.800 0.700       10                     0.5                   0.700
    S003 0.500 0.125        8                     0.0                   0.125
    S004 0.429 0.286       14                     0.0                   0.286
 pairwise_covariance_lower pairwise_covariance_upper
                  -0.00274                   0.00960
                  -0.00600                   0.01400
                  -0.00781                   0.00781
                  -0.00875                   0.01166

projected_scenario_role: stress_scenario_not_bound
```

C.8. **Variance smoothing (experimental).** The opt-in smoother applied to the Scenario-A table.

```
est_sm <- sm_smooth_variance(est, method = "loglinear")
```

```
 site_id  n    se se_smoothed  var_method
    S001  9 0.167       0.167 arcsine_vst
    S002 10 0.158       0.158 arcsine_vst
    S003  8 0.177       0.177 arcsine_vst
```

On the arcsine scale the canonical standard errors are already deterministic in n, so the smoother reproduces them; this illustrates why the fixed-seed audit found no basis for promoting smoothing to a default (Appendix D).

C.9. **Downstream empirical-Bayes demonstration.** The six-line EB step of the main text, with the summary quantities it prints.

```
d <- as.data.frame(est)                       # transport, not validation
d <- d[is.finite(d$theta_hat) & is.finite(d$se) & d$se > 0, ]
tau2 <- max(0, var(d$theta_hat) - mean(d$se^2))   # method of moments
mu   <- weighted.mean(d$theta_hat, 1 / (tau2 + d$se^2))
lam  <- tau2 / (tau2 + d$se^2)
d$theta_eb <- mu + lam * (d$theta_hat - mu)
d$p_raw <- sin(d$theta_hat)^2; d$p_eb <- sin(d$theta_eb)^2
```

```
tau^2 = 0.0206 | mu (arcsine) = 0.715 | shrinkage weight: min 0.40 / max 0.89
Spearman correlation of raw vs EB ranks: 0.988
Sites in the raw top 10 that stay in the EB top 10: 8 of 10
```

```
largest rank movements:
 site_id  n p_raw p_eb rank_raw rank_eb
    S001  9  0.11 0.28       50      45
    S031 12  0.58 0.51        8      13
    S012 12  0.33 0.38       37      33

Top-ranked site S005 (p = 0.88): its 95% interval overlaps those of 23 of
the other 49 sites
Median site-year enrollment in 2024: 24
```

Appendix D. Audit, Suppression, Fréchet, and Smoothing Semantics

D.1. **Three diagnostic levels and the severity matrix.** `sm_diagnose()` returns S3-classed tibbles at three levels: `summary` (one row per object; the 1×40 tibble of Appendix C, covering cell/group/indicator counts, flag tallies, identification status counts, scalar-validity facts, covariance presence and validity, scale compatibility, suppression-sensitivity role, smoothing facts, and staleness), `row` (one row per cell with flags and a severity tier), and `vcov` (one row per covariance carrier with eigenvalue margins and scale-compatibility facts). We classify findings on an *error* / *warning* / *note* / *ok* matrix; for example, an exact census with zero sampling uncertainty is a *note* (a legitimate state), while an estimate paired with a stale or scale-incompatible covariance is an error.

D.2. **Publisher suppression.** `sm_suppression_report()` audits aggregate inputs before estimation. At estimation time the default policy (`"drop"`) preserves suppressed cells as missing-value audit rows (keys and flags are retained and no numbers are fabricated), distinguished by `estimate_status = "suppressed_missing"`. The legacy worst-case mode (`"upper_bound"`) is intentionally demanding to invoke: it requires `suppression_sensitivity_acknowledge = TRUE`, stores the Bernoulli worst-case variance ($p = 0.5$, variance $0.25/n$ when the denominator is observed) only in separate `sensitivity_*` columns with `estimate_status = "suppression_sensitivity"`, and refuses to fabricate a numeric variance when the denominator itself is hidden. Sensitivity rows are excluded from ordinary covariance construction, from Fréchet analysis, and from smoothing; they are stress scenarios and are not treated as estimates.

D.3. **Formal Fréchet intervals versus projected stress scenarios.** For D1 inputs, `sm_frechet_envelope()` separates what is provable from what is exploratory.

For the formal component, when the declared provenance is `same_units` with a common denominator per site-year, IID plug-in variances, and no finite-population or bias correction, the joint proportion of each indicator pair is sharply bounded by the Fréchet–Hoeffding inequalities (Fréchet, 1951; Hoeffding, 1940; Nelsen, 2006),

$$\max(0,\, p_k + p_l - 1) \;\leq\; p_{kl} \;\leq\; \min(p_k,\, p_l), \tag{D.1}$$

which induce exact endpoints for the pairwise covariance, $(q^{\pm} - p_k p_l)/n$, and correlation. We return these in `raw_pairwise_intervals` with `interval_scope = "formal_raw_pairwise_interval"`; the required `population_regime = "d1a"` argument fails closed when the provenance does not support the claim.

Turning to the stress scenarios, for $K > 2$ the assembly of all pairwise lower (or upper) corners into one matrix generally leaves the positive-semidefinite cone, and projecting back (Higham, 2002) can move individual entries outside their own pairwise intervals. The projected matrices are therefore labeled `projected_negative_dependence_stress` or `projected_positive_dependence_stress`, with role `stress_scenario_not_bound`; sign changes, order reversals, and interval violations are reported in long-format projection diagnostics, and the projections are never returned as ordinary covariance carriers. The recommended use is to rerun a downstream analysis once per scenario; the pair should not be treated as bounds. Under `"different_units"` or `"unknown"` provenance the analysis is available only as an explicitly acknowledged heuristic.

D.4. **The experimental status of variance smoothing.** In a pre-registered, fixed-seed simulation audit (60 replicates; denominators 8–150; true rates 0.02–0.98; small-/large $\times$ boundary/interior strata), we compared unsmoothed standard errors against log-linear and generalized-additive smoothers (Wood, 2017) on both reported scales, with promotion criteria fixed in advance (variance-MSE improvement, worst-stratum guardrails, coverage error, inverse-variance-weight stability). No candidate met the joint criteria (raw-scale smoothing roughly doubled the relative variance MSE), and `sm_smooth_variance()` therefore remains opt-in, append-only, and labeled experimental: a generalized variance-function sensitivity tool that makes no area-level model claim. Canonical standard errors are never replaced automatically, and overwriting is refused whenever a matching-scale covariance would be left stale.

Appendix E. Downstream Recipes

The package does not favor any particular consumer; in this appendix we record the recipes used in the main text, together with pointers for common destinations.

E.1. **Normal–normal empirical Bayes in base R.** With reported pairs $(\hat{\theta}_j, s_j)$ on the arcsine scale (so s_j is free of the unknown rate), the classical normal–normal model (Efron, 2016; Morris, 1983) is $\hat{\theta}_j \mid \theta_j \sim N(\theta_j, s_j^2)$, $\theta_j \sim N(\mu, \tau^2)$. A method-of-moments fit is

$$\hat{\tau}^2 = \max\Big\{0,\ \mathrm{Var}(\hat{\theta}) - \overline{s^2}\Big\}, \qquad \hat{\mu} = \frac{\sum_j \hat{\theta}_j/(\hat{\tau}^2 + s_j^2)}{\sum_j 1/(\hat{\tau}^2 + s_j^2)}, \tag{E.1}$$

with posterior means $\hat{\mu} + \lambda_j(\hat{\theta}_j - \hat{\mu})$, $\lambda_j = \hat{\tau}^2/(\hat{\tau}^2 + s_j^2)$, posterior variances $\lambda_j s_j^2$, and back-transformation $p = \sin^2(\theta)$ for reporting. This is the demonstration in the main text, with the back-transform added for reporting; production analyses can swap in nonparametric deconvolution (REBayes; Koenker and Gu 2017), robust EB intervals (ebci; Armstrong et al. 2022), report-card procedures (Kline et al., 2024; Walters, 2024), random-effects meta-analysis (metafor; Viechtbauer 2010), or hierarchical models (Bates et al., 2015) without changing the first stage.

E.2. **Joint contrasts without inversion.** For a within-site contrast $a^\top\hat{\boldsymbol{\theta}}$ across indicators (e.g., FRPM minus SNAP), the delta-method variance is $a^\top V a$ with V the site's covariance carrier on the matching scale. The documented pattern evaluates the quadratic form directly, without inverting V, so rank-deficient multinomial matrices and exact-census zero matrices remain consumable.

E.3. **Cautions.** Three practices the documentation explicitly warns against: pairing reported-scale estimates with the raw-scale matrix (scale mismatch is a diagnosed error); giving suppressed or sensitivity rows inverse-variance weight (they are retained as labeled states and receive no weight); and treating projected Fréchet stress matrices as bounds or as ordinary covariance inputs.